

measure that behaves well on “nice” spaces

- **Why do we care?**

#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

Radon Measures

- A Radon measure is a **measure that behaves well on “nice” spaces**, specifically:
 - It's locally finite: gives finite mass to some neighborhood of every point in the space
 - It's *inner regular*: can be approximated from inside by compact set. Formally, for a topological space $\mathcal{X}$: $\mu(A) = \sup\{\mu(K) \mid K \subset A, K \text{ compact}\}$
- **Why do we care?**
 - Most probability measures on $\mathbb{R}^n$ are Radon
 - Allows integration, pushforwards, and duality to work smoothly
 - Ensures OT problems are well-posed on general spaces
 - Many theorems (Prokhorov's theorem, weak convergence) phrased in terms of Radon measures
- The set of Radon measures on $\mathcal{X}$ is denoted $\mathcal{M}(\mathcal{X})$
- $\mathcal{M}_+(\mathcal{X})$: non-negative Radon measures. $\mathcal{M}_+^1(\mathcal{X})$ (or $\mathcal{P}(X)$): Radon probability measures.

Measure Landscape